

Yeast scRNA-seq methods: what separates the cells, and which molecule carries the cell barcode

A redrawing of Fig. 1 of Nadal-Ribelles 2024, reorganized around the one question that separates the library types. Every molecule is drawn in a single fixed orientation, 5' left to 3' right with P5 always on the left, and colors are those of Figs. 2 and 3.

a What keeps one cell's molecules separate from another's

cells profiled per experiment

Six options, ordered by throughput. The last one is not an isolation method at all.

Microfluidic chip (C1)	FACS into plates	Micro-dissection	Droplet (10x, Drop-seq)	Microwell array	Combinatorial indexing
up to 96 per chip one chamber per cell	~285 cells one well per cell	~2,000 cells one well per cell	6,000–100,000 an oil-walled droplet 152,000 preindexed (human; Fig. 4)	up to 1,061,865 a printed microwell	25,000–240,000 per run nothing: the fixed cell is its own container
per-cell phenotype recorded (imaging, reporters, sorting gates)			no per-cell phenotype; cells are pooled and read out only by sequence		

b The question that separates the library types: which molecule carries the cell barcode?

Four answers. The original figure draws the products; this draws the choice that produces them.

the oligo-dT	the TSO	the Tn5 adapter	nothing in the molecule
CB UMI dT	CB UMI rGrGrG	Tn5 CB	the well is the barcode
3' end libraries. Barcode is on the primer, so it lands next to the poly(A) end. Droplet, microwell, combinatorial indexing.	5' end libraries. Barcode rides the template-switch oligo, so it lands at the transcript's 5' end. Commercial droplet.	5' end, plate route. Neither oligo is barcoded; identity is added later, at tagmentation. yscRNA-seq, the only 5' method run in yeast.	Full-length. Tn5 carries universal adapters; the cell is identified by which well the library came from. SMART-seq2, plate only.

c Where the tags end up – all three drawn in one orientation, on one column grid, so they can be compared

3' end	5'	P5	Read 1	CB	UMI	cDNA – the 3' end of the transcript	Read 2	index	P7	3'
5' end	5'	P5	Read 1	CB	UMI	cDNA – the 5' end of the transcript	Read 2	index	P7	3'
full-length	5'	P5	Read 1	no cell barcode		cDNA fragment – tiles the whole transcript	Read 2	index	P7	3'
in 3' and 5' libraries Read 1 carries the cell barcode						this index is the cell identity				

Read panel b and panel c together and the taxonomy collapses to one rule: a cell barcode is only needed where cells are pooled before sequencing. Droplet, microwell and combinatorial methods pool early, so identity has to be written into the molecule, and it goes wherever the barcoded oligo sits – the poly-dT end for 3' libraries, the template-switch end for 5'. Plate methods never pool until the very end, so the molecule needs no barcode at all and the Illumina sample index does the work, one index per well. That is the same accounting split-pool uses when its sublibrary index becomes barcode round 4.

Segments	transcript / cDNA	cell barcode (CB)	UMI	TSO rGrGrG	container / enzyme	handle / adapter	Illumina sample index	red dashed = absent, or doing an unexpected job
----------	-------------------	-------------------	-----	------------	--------------------	------------------	-----------------------	---